

```
df = pd.read_csv("heart.csv")

print("Dataset Shape:", df.shape)
print(df.head())
```

Dataset Shape: (1025, 14)

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	\
0	52	1	0	125	212	0	1	168	0	1.0	2	
1	53	1	0	140	203	1	0	155	1	3.1	0	
2	70	1	0	145	174	0	1	125	1	2.6	0	
3	61	1	0	148	203	0	1	161	0	0.0	2	
4	62	0	0	138	294	1	1	106	0	1.9	1	

	ca	thal	target
0	2	3	0
1	0	3	0
2	0	3	0
3	1	3	0
4	3	2	0

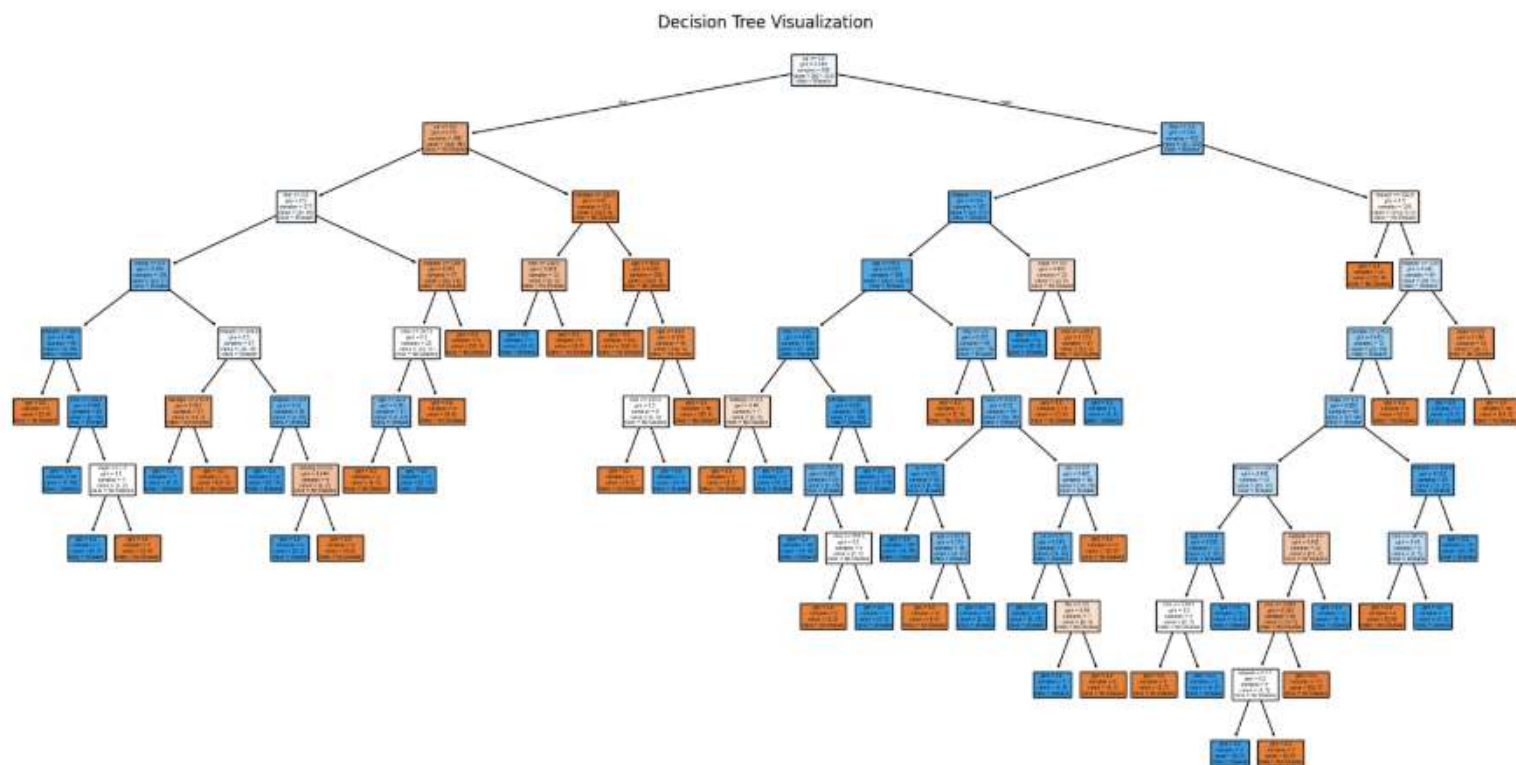
```
print("\n=== Decision Tree Accuracy ===")
print(accuracy_score(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt))
```

=== Decision Tree Accuracy ===

0.9853658536585366

	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205

```
# 5. VISUALIZE DECISION TREE
# =====
plt.figure(figsize=(20,10))
plot_tree(dt, feature_names=X.columns, class_names=["No Disease", "Disease"], filled=True)
plt.title("Decision Tree Visualization")
plt.show()
```

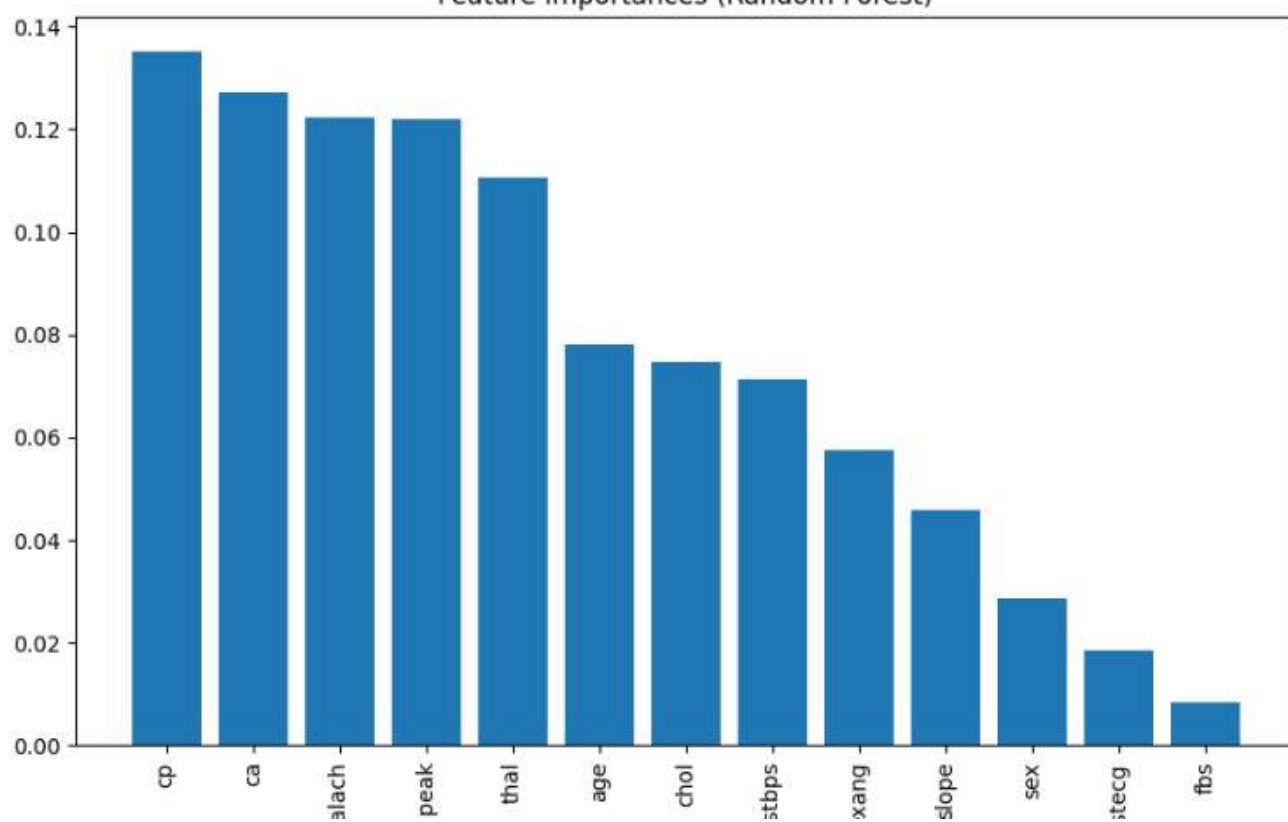


```
print("\n=== Random Forest Accuracy ===")
print(accuracy_score(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
```

```
=== Random Forest Accuracy ===
0.9853658536585366
```

	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205

Feature Importances (Random Forest)



```
print("\n=== Cross Validation Results ===")
print("Decision Tree CV Accuracy:", dt_scores.mean())
print("Random Forest CV Accuracy:", rf_scores.mean())
```

```
=== Cross Validation Results ===
Decision Tree CV Accuracy: 1.0
Random Forest CV Accuracy: 0.9970731707317073
```
